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Continuous Innovation in Hard Rock Reservoirs: Trials of Vibration Eliminator, Brine-Based Mud and MSF Completion to Optimize Time and Cost in Northern Oman Gas Fields

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Abstract

Petroleum Development Oman is one of the leading companies in Sultanate of Oman and the region, known by overcoming the oil & gas challenges with strategic and incremental steps by adopting the latest advanced solutions for maximized sustainable outcomes for the company and the nation.

This paper outline PDO trials to optimize time and cost in Northern Oman gas fields, which are combination of drilling and completion approaches. These trials were customized to fit the specific requirements of the hard rock reservoirs for improved cost and time, that are both reflected in lower GHG and carbon footprints, where those factors are main pillars of PDO's 2030 & Oman's 2040 visions.

Trials of vibration eliminating tool, brine-based mud & MSF (Multi-stage fracturing) completion were planned with clear key performance indicators and goals, which are constructed, monitored, and presented to demonstrate the previous challenges and present clearly what would be the acquired improvements from these trials. Overall outstanding improvements have been achieved, resulting in a major reduction in drilling days (up to 50% reduction in time consumption) and lower utilization of expensive service units for post-drilling activities like coiled tubing and wireline units (about 60% reduction). This translated PDO's strategy towards innovation, cost consciousness, and the mindset of value maximizing.

Introduction

Petroleum Development Oman is one of the major operating companies in the Sultanate of Oman, covering 30% of Oman's concession areas from the northern to southern ends of the nation. PDO has a wide spectrum of operations throughout the well's lifecycle, from exploration to abandonment, across a variety of hydrocarbon fields (oil, condensate and gas) and with different classifications of risk rating from very low to very high-risk wells.

One-third of the well's duration in Oman's Northern gas development fields is spent on drilling the deep sand reservoir that is similar to Middle East operators' fact about consuming 30-40% of well duration in drilling the deeper sections as per (A Albattashi et al. 2020). Drilling hard rock is associated with long drilling duration, high temperatures, tool failure, bit dulling, lost in hole (A. Bourgoyne et al. 2017).

Thus, any minor improvement in performance, especially in this reservoir section, would represent a major positive impact.

In PDO's well engineering directorate, one of the main priorities within gas development fields is delivering quality wells with minimized delivery days, cost efficient, and high integrity design for extended production operations. This is all translated into a lower carbon footprint & lower GHG, which are main parts of the company's organizational purpose and Oman's 2040 vision (Austin Matthews et al. 2020).

In Northern Oman, persistent hard drilling challenges in the reservoir interval that are mentioned previously, have historically delayed project timelines, intensified operational costs, and increased the likelihood of complicated downhole challenges. So, this paper explores our company's continuous efforts and the spectrum of recent trials aimed at mitigating these challenges through innovative drilling fluid systems, vibration eliminators, and advanced completion components.

The significance of these trials lies in the pre-collection of reliable data from previous PDO's operations, flawless execution due to limited trials, the wide variety of fields, a proper selection of candidates, and the quality of real-time data collected for a robust post-trial analysis and replication across PDO's large fields.

This paper is structured into several sections, beginning with an overview of the challenges encountered. It then details the candidate wells selection process and the criteria used, supported by relevant data samples. The execution and planning of the trials are subsequently discussed, along with an analysis of the challenges and reliability issues faced during and after the trials. Finally, the paper concludes with a summary of the outcomes and highlights the benefits achieved.

Challenges Overview

Drilling the reservoir section in Oman's North fields can reach depths exceeding 5000m TVD, highlighting the significant challenges encountered during the drilling, completion and production phases.

In drilling, the reservoir sections tend to consume 60% of the wells duration where we utilize 4 to 5 bits to reach Total Depth (TD). Drilling for extended durations tend to increase the challenges of well placement due to well drifting (because in planning phase we plan and simulate the bottom hole assembly components to achieve drifting resistance and to drill till final depth without compromising the drilling rig capacity and limitations of over pulling safety factors, both of those factors are contradicting each other so we plan to balance between both factors), drilling string components failure like washouts and twist off if fatigue and subsurface stresses are not monitored precisely, and these failures had been experienced in PDO where a highly time consuming pull out of the string were done to investigate and rectify downhole washouts.

Also, long durations accompanied with high downhole temperature are a reason for raising the drilling fluids temperature leading to drilling fluid properties deterioration, thus, high consumption of chemicals used for dilution and treatment, extensive usage of expensive cooling systems and most importantly the re-crushing of the cuttings and the increased percentage of Low Gravity Solids (LGS) in the fluid system. This is leading to compromise drilling efficiency due to lost energy at bit to remove the cuttings up the annulus away from the bit cutting structure. LGS significantly reduces rate of penetration and usually is responsible for numerous electronic failures within the MWD packages. Due to the high temperatures, fluid deterioration was observed even during static conditions. As a result, costly, highly engineered high-temperature mud formulations were required whenever a trip was planned.

Bits vibrations and un-delivered energy to bits are major and persisting challenges, where the full parameters are applied but most of the energy is lost in lateral, tortional and vertical vibrations. There were intensive efforts from PDO's planning & design team to enhance bits designs, Botton Hole Assembly (BHA) components and drilling roadmaps, but still the challenge is present. Vibrations create very complicated indirect impacts on the efficiency and durability of booster motors which pre-maturely fail due to excessive vibrations and un-delivered energy. Vibrations also wear out the bit and sequentially impact the other components of the BHA such as the MWD components but also affects the whole drilling system including the Top Drive System (TDS) on surface.

Drilling this challenging reservoir section presents a series of interrelated issues that occur sequentially or simultaneously, making strategic decision-making extremely complex. Motor failures are common due to the previously discussed challenges, and these failures can escalate into more serious complications such as twist-offs or lost tools in the hole. In such cases, complex fishing operations at deep intervals are often required (Figure 1). When retrieval is not possible, time-consuming sidetracking becomes the only viable option. As a result, a recurring and critical question arises: *Should we stop drilling and pull out to avoid further complications, or should we continue and risk escalating the situation?*

Figure 1—Twisted motor fishing operation.

In term of post drilling phases, which are completion, hydraulic fracturing and production, this section is mainly targeting 2 different reservoirs with different pore pressures, so the typical method is completing the well with mono-bore cemented 4.5" tubulars, then the process continues to an extended plug & perf method taking many days for perforating, plugging with copper head plugs and milling. This traditional method has many drawbacks such as:

1. Low precision of perforations due to perforating tool length limitations and closely grouped units of paying and non-paying zones.
2. Extended duration and holding of expensive resources on location for Plug & Perf, this underutilization includes coiled tubing units along with wireline to be in standby mode for each other while performing plugging, perforating and fracturing (Mahrouqi et al. 2020).
3. Conventional milling of copper head plugs includes very complicated milling procedures, occasionally, escalated to tubular damage.
4. Hydraulic fracturing pressure and proppant penetration into the least competent perforation withing a single unit leading to non-fractures paying zones and screen-out of perforations.
5. Since the well is producing from 2 different reservoirs, there are Gas-water contact layers at each reservoir, and there are multiple reasons for shutting off zones, but this compromises major parts of the reservoir if a shallow shut off is required.

The challenges outlined above contribute significantly to increasing operational time, project delays and inflated Capital Expenditure (CAPEX). To overcome these challenges, a strategic shift towards advanced, integrated drilling and completion technologies is essential. This includes the adoption of optimized bit and BHA designs, real-time vibration monitoring, and thermally stable drilling fluids, and the deployment of modern multi-stage fracturing systems that provide enhanced precision, efficiency, and reservoir control. By focusing on these critical areas, operators can improve operational performance, minimize non-productive time, and unlock the full production potential of these complex reservoirs.

Preparations of candidate wells & data collection

Candidate wells selection process played a fundamental part in the success of the trials. PDO planning & design team along with operations and subsurface team collaboratively set the characteristics of the candidate wells which are:

- Having clear objectives, and well-defined Key Performance Indicators (KPIs). Examples of Key performance indicators are listed in Table 1: number of runs to drill the section, Rate Of Penetration (ROP m/hour), section duration in days, fish-free section, Issues free section related to trials factors.
- Measurability and evaluation of the candidate wells is well defined prior to the trials to set the reference criteria for an ongoing benchmarking and ongoing decision-making method while the trials are being performed.
- Controlled variables and calculated risk are taken into account as unexpected external factors continue to increase the likelihood of failures and unanticipated outcomes. All risks are formally registered and strictly mitigated to ensure that every aspect of the trial aspects remain within PDO's approved risk matrix and acceptable tolerance levels.
- Scalability. The primary objective of these trials is to evaluate their potential for replication across other applicable sections/fields. Every worth replicating practice is being customized and replicated for maximized positive results.
- A critical factor in choosing candidates well is performing trials in very common formations which consist of hard rock characteristics, which are applicable and available in many fields, faults and losses free for logical benchmarking and for the reason of completing trials from end to end.
- For MSF completion, the criteria were to run this very expensive completion in problem-free section, equipped with motorized shoe to ensure reaching the targeted depth. The selected candidates should also contain multiple pay zones to accommodate the installation of more than 12 sleeves of MSF for better utilization, relevancy and to measure the success of placing more than 10 sleeves precisely in an unevenly distributed pay zones.
- In addition to the above, commercial evaluation is a critical criterion for assessing the economic viability and operational value of the new technology. These trials underwent a thorough commercial assessment to ensure they delivered added value without increasing overall well construction cost.

Table 1—Example of drilling data collection and targets.

Data collection	Reasons	Targets
Formation specific Rate of penetration ROP (m/hr)	To define and distinguish ROP/formation for bit road map design and gaining ROP in softer formation.	To achieve 30% improvement above field best composite time.
Average Rate of penetration ROP (m/hr)	Define the average ROP to be achieved across all section parts.	To achieve 30% improvement above field best composite time.
Section duration (days)	This illustrates overall section durations including round trips, back reaming, subsurface challenges, losses curing and fishing operations.	Derived from ROP improvements and issues-free goal. Comprehensive goal was to reduce section duration by 10 days.
On bit hours (hrs)	This distinguishes solely the bit energy and work on bottom, in combination with drilling fluid design to drill hard formations.	Derived from ROP performance.
NPT (hrs)	Nonproductive time is an illustration of negative impacts produced directly and in-directly by trials.	To drill the section with 0 NPT hours related to trials.
Number of runs	Illustration of number of bits used, and how good is the integration of all systems for a durable run.	To drill the section in 2 runs only.
Applied drilling parameters	Defining the parameters used on each lithology, and to know what gaps to be addressed in our trials since all rheology and BHA are changed.	To maximize parameters to the upper limit over the whole section, by enhancing BHA design & weight.

Execution of trials

Vibration Eliminator Overview

Drilling vibration is a common and critical challenge in oil and gas well construction, especially in vertical wells. Uncontrolled vibrations can lead to a range of issues including equipment failure, poor drilling performance and increased non-productive time (NPT). This tool is one of the new technologies to limit High Frequency Torsional Oscillation (HFTO), reduce stick/slip occurrences, and improve bit durability. It is part of the drilling BHA which acts as oscillations shocks dampening and its position is to be as close as possible to the bit, which is believed to be the main source of vibration. It is also equipped with 2 vibration sensors on top of each other to record and monitor the dampening effect during the run. The lower sensor helps by providing the actual downhole vibrations close to the bit and the upper sensor reads the dampening effect. The post run analysis can be done to reflect the vibration reduction. Its value has been seen clearly in ROP enhancement & extending bit life in hard rock drilling avoiding frequent trips in long drilled sections.

Vibration eliminator tool Planning & Execution

The right applications of this tool are in hard rock drilling with interbedded formations that promote stick-slip behavior, directional and horizontal wells, where torsional instability is more pronounced and scenarios requiring maximum drilling parameters without over-torqued connections or fatigue failures. Also, it can be deployed in long open hole sections such as deep gas wells where 2-3 trips are required to completion the section. The planning started by engaging with the service provider to understand the tools features, its potential and the value added to enhance drilling performance. Then, detailed technical review has been done to evaluate its compatibility with other components on bottom hole assembly and its operational limits. Furthermore, proper placement of the tool in the Bottom Hole Assembly (BHA) is critical for maximizing its effectiveness and it is usually directly above the Bit (Motor or RSS Assembly). After detailed planning, vibration eliminator tool has been deployed in one of the deep gas fields in PDO in 2 different wells.

Brine-Based Mud Overview

A Brine-Based mud system is a type of drilling fluid where the base fluid is a brine solution—typically a water-soluble salt such as sodium chloride (NaCl), potassium chloride (KCl), calcium chloride ($CaCl_2$) and polymers. This system is primarily used in drilling, completion, and workover operations, especially in high-pressure high temperature (HPHT). Brine-based mud systems can be engineered not only for wellbore stability and formation protection but also to enhance the Rate of Penetration (ROP) during drilling. This is achieved through a combination of optimized mud chemistry, lower solids content, and formation-friendly properties. PDO has used this system in hard rock drilling applications to enhance rate of penetration while maintaining hole stability and ensuring safe & efficient operations. Brine Based Mud has low viscosity, low yield point (YP) & low (low gravity solids) LGS%. Having this formulation with thin properties, it improves the jet impact force at the formation & hence improves the rate of penetration. Low rheology mud would introduce the risk of hole cleaning however this is compensated by having high flow rate which produces turbulent flow regime in the annulus generating higher energy to lift the cutting to surface.

Brine-Based Mud Planning & Execution

The planning starts with the selection of the right well's candidate based on mud weight range as it only works in a certain range of mud weight. Detailed simulation of hydraulics to ensure high flow rate with turbulence flow regime is achieved in all the time during drilling. Also, hydraulics simulation estimates the highest potential circulating pressure during drilling and compares it with the rig pump capacity. Proper follow up during the mixing of the mud to achieve the right parameters & maintaining them as per the drilling fluid program during drilling. The success factor of this mud is to keep low solids % to maintain low YP and achieve the required jet impact force & turbulent flow for hole cleaning.

MSF Overview

The MSF (Multi-Stage Fracturing) technology involves deploying downhole Multi-Stage Fracturing tools along with the completion string—such as sleeves, open hole packers, and open hole anchors—within a specified open hole size, serving as an alternative to the conventional Plug & Perf hydraulic fracturing system.

The (MSF) system offers several advantages that contribute to the optimization of fracture deployment. It introduces cost and time savings by minimizing the need for complex interventions, thus enhancing operational efficiency and enabling the delivery of more wells using the same resources due to faster fracturing operations. The system also facilitates earlier well hook-up, leading to accelerated production. By eliminating or significantly reducing interventions such as E-line perforation, plug setting, and coiled tubing milling, MSF ensures access to deformed intervals while minimizing flaring and Green House Gas (GHG) emissions. Additionally, the ability to perform un-sequential fracturing allows for stress shadowing relief, reducing the risk of screen-outs. The use of reclose able sleeves further enhances flexibility by enabling selective production and providing means for water shut-off when required.

MSF system component consists of toe Kick Start sleeve, Drop Shift sleeves (single entry & limited entry) and cementing accessories such as float shoe, float collar, special landing collar and special wiper dart plug (Figure 2).

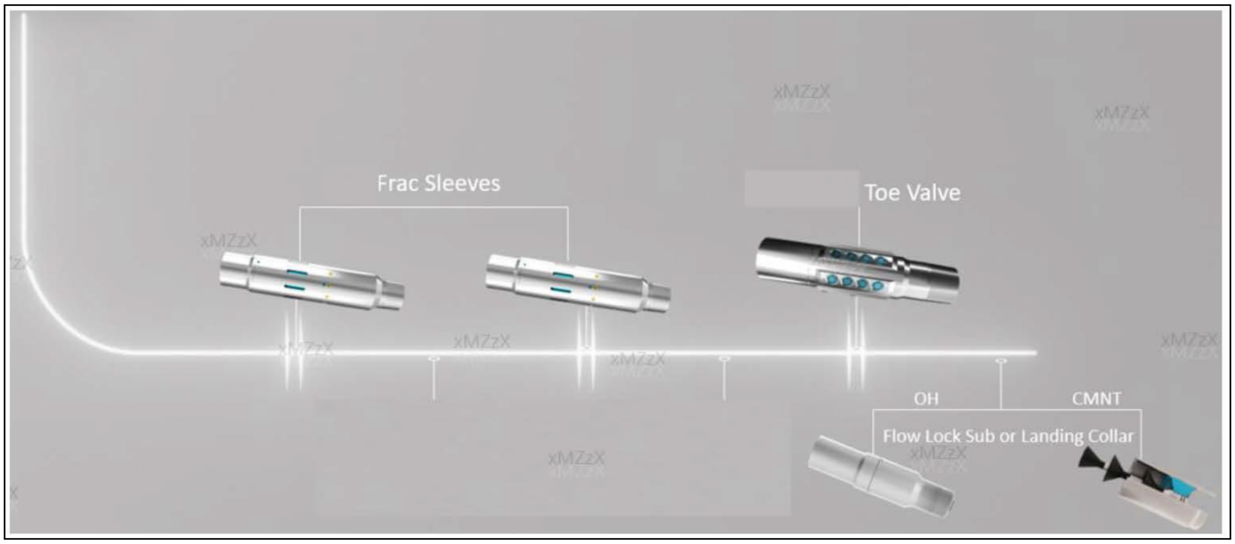

Figure 2—MSF completion components. (Rodoplu et al. 2024)

MSF Planning & Execution

The planning of the MSF deployment started first from selecting the right well candidate which has multiple producing zones and where several stages are required to do hydraulic fracking. After the targeted reservoir was drilled to Total Depth (TD) and the open hole evaluation conducted, the desired sleeve depth and type were selected. A precise tally was generated to place these sleeves against interested zone. Once completion was deployed to well TD, a wireline CCL correlation (Casing Collar Locator logging) is conducted to ensure sleeves placement as per the desired depth against open hole GR signature (Gamma Rays emitted by formation particles). This step is to confirm tally accuracy and cater for thermal & mechanical elongation of the completion. Then, before cementing operation, WellCat model was run to ensure the circulation pressure is not exceeding the toe kick start valve opening pressure (pressure activated toe valve). After the well was cemented successfully, frac operation started with completion integrity test followed with opening the toe kick start valve to establish communication with the formation and start fracking stages by dropping the balls to open the sleeves against the interested zone. The balls with ball seat work as an isolation stage between stages. E-line runs, perforation operation and milling plugs were eliminated.

Trials Challenges and reliability

Vibration Eliminator

The tool was the first trial of its kind, and therefore, no historical data was available to support its deployment. Additionally, due to previous major incidents involving twist-offs and tools lost in the hole, there was a general reluctance from management to approve its use in deep wells. Compounding the challenge, the downhole sensors were not capable of providing real-time data, which prevented timely alterations or modifications to drilling parameters. Moreover, the implementation of the tool required a dedicated drilling run to initially drill the stage tool, float collar, float shoe, and shoe track, ensuring the bit and tools remained undamaged. This approach conflicted with the company's standard practice of drilling the entire section in a single run, as the need for a separate operation to drill plugs and the shoe track would have resulted in additional time and cost, rendering the method irrelevant to current operational efficiency goals.

Brine-Based mud

Intensive modeling of hydraulics, Equivalent Circulating Pressures (ECP) and the rig's pumping pressures were carried out through detailed simulations to support the use of a highly engineered mud fluid. This fluid was designed to possess low rheology and promote turbulent flow, enhancing drilling performance. During the planning phase, the mud system underwent rigorous evaluation through the well control panel to ensure its ability to generate the necessary hydrostatic head and form a competent filter cake in depleted formations. This process was critical to ensure the fluid's compatibility with PDO's well control standards. The Brine-Based mud system came after the drilling of a highly shaly formation using oil-based mud (OBM) in the previous section and presented additional challenges. Tanks had to be thoroughly cleaned to eliminate any residual OBM or cement contaminants, as the brine-based system required a clean, oil-free environment. Moreover, preparing this system involved mixing large volumes of chemicals and sourcing very fine shaker screens (above API 320 grade) from an international vendor, as required for the low solids mud system. During drilling operations, maintaining consistent mud rheology within the designed parameters proved difficult due to fluctuations in yield point (YP) and plastic viscosity (PV); primarily caused by shaker screen damage and solids recirculation. This issue was managed by frequent dilution of the mud and re-dressing the shakers with new screens.

MSF Completion

MSF completion contains overly expensive and highly precision components, which required well supervised transportation and high precision make-up in the vendor's workshop, due to its special design and material. The rig handling and make-up of MSF items were complicated in terms of the need of special elevators and make-up tongs. In terms of subsurface, MSF tally was designed to be within 0.5m tolerance in the presence of temperature & mechanical elongations, circulating and cementing up- lift effect and in combination of highly unprecise production zones which required detailed and complex tally to fit the sleeves on the right interval with all mentioned consideration., As an example, 32 pup joint were used in one of the MSF tallies. After installation of the MSF completion, correlation log was done to confirm the placement of sleeves before performing the final step which is cementation. This is a combination of sub surface operations and wireline engineer verification.

Trials Outcomes

As highlighted in the abstract of this paper, PDO aims to address oil and gas challenges through strategic and incremental step changes by adopting advanced technologies to maximize sustainable outcomes for both the company and the nation. Key milestones in these initiatives were successfully achieved through extensive planning and outstanding execution. The following is a summary of the trial's outcomes:

In short, Vibration Eliminator is a powerful tool for operating companies aiming to boost drilling efficiency and reliability by proactively controlling vibrations. Especially in persistent stick-slip or looking to push the limits of drilling performance while protecting drilling tools. PDO's outcomes in Two different trials were 11% improvement in Rate of Penetration, 16% higher in drilling interval meterage with the single bit and successfully eliminated 10-27% shocks (low & high levels) as illustrated in Figure 3.

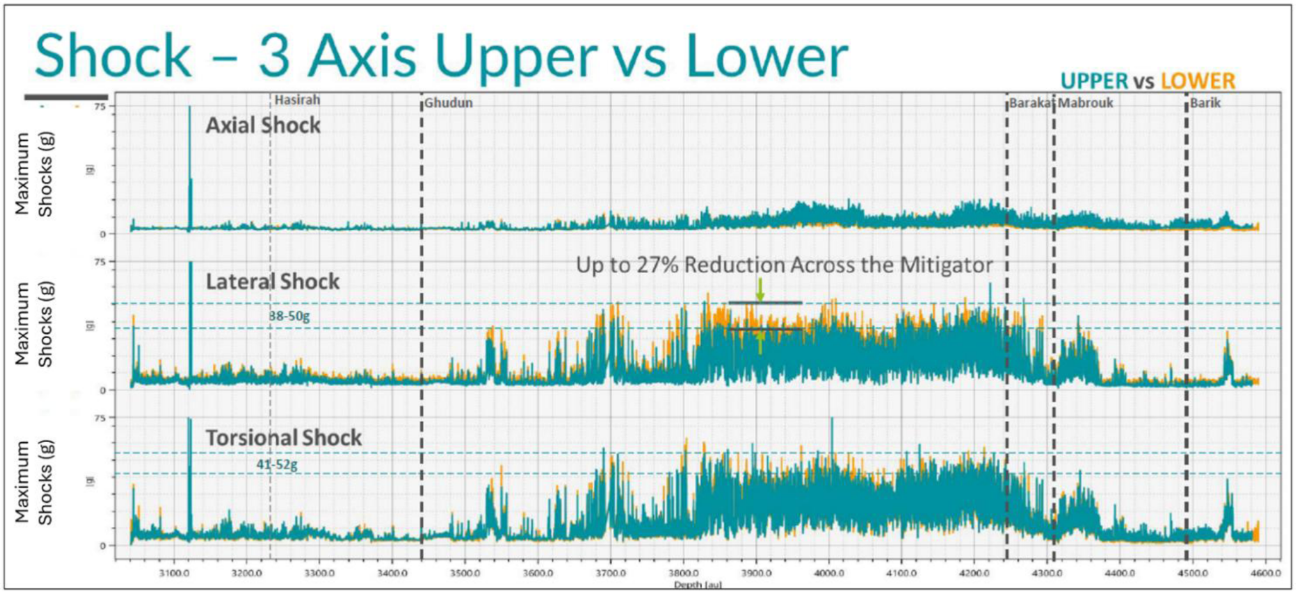

Figure 3—Tool's Upper and Lower Vibration Sensors Readings.

For Brine Based Mud, three trials were completed in 3 different fields within deep gas fields with all hard rock application & long open hole section. These trials yielded the following key results (Figure 4 illustrates one of the trials' outcomes):

- The objectives were achieved, validating Brine Based Mud efficiency in improving drilling performance by reducing operation time and cost.
- Recorded ROP improved by 40-50% from the best ROP in offset wells and 80% higher than field average. The ROP per formation achieved can be seen in figure 4.
- 30-50% direct reduction in hard rock drilling section time.
- Achieved zero NPT related to mud issues (e.g., stuck pipe, hole instability and open hole logging issues).

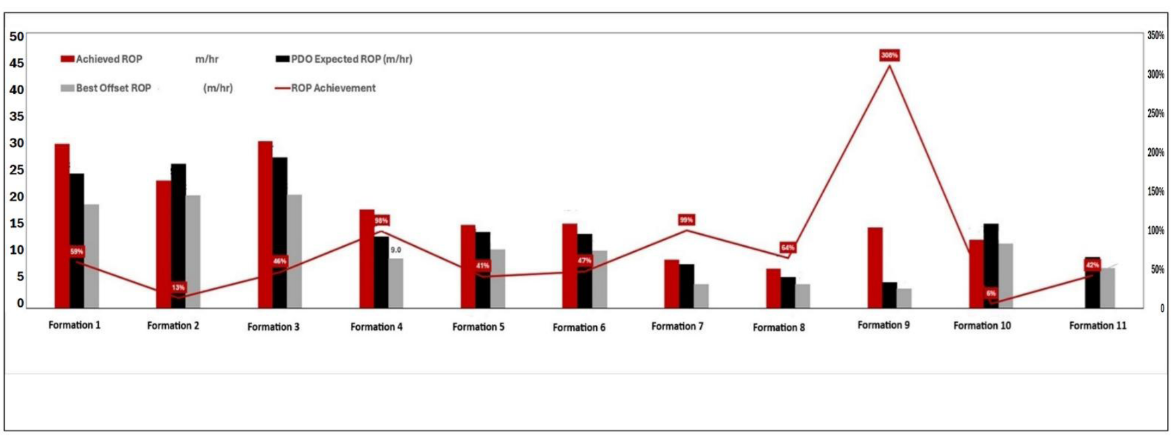

Figure 4—Rate of Penetration % Improvements per Formation.

For MSF completion, 6 Wells of Multi-Stage Fracturing of Mono-bore cemented MSF were successfully deployed with Vendor A & Vendor B with both 15000 &18000 PSI rated MSF systems in 2024 and 2025 in different deep gas fields. Teflon gauge cutter run was performed before rig move to confirm Held Up

Depth (HUD) of the wells, which validated the success of installation. Inhibited brine was used to displace cement, which eliminated the risk of prematurely activating kick start valve during cementation.

The toe kick start valve was successfully activated with surface pressure as per calculation sheet after Cement Bond Logs (CBL) and tubing integrity test, which eliminated the cost and time of Wireline intervention in conventional plug and perforation system.

Total of 4 wells already were hydraulically fracked, and pre-set objectives were successfully achieved. The actual saving was freeing almost 60% of resources needed for Frac operations along with 20%-30% Cost improvement and 30-40% Time reduction. Furthermore, all the complicated milling of Frac plugs & cleaning-out runs are now eliminated.

Conclusion

In conclusion, the trials and technologies outlined in this paper demonstrate significant potential for replication across similar applications in the oil and gas industry, particularly in fields characterized by hard and abrasive formations, also by ideas integration such as combining Brine-Based Mud with the vibration eliminating technology for maximized benefits. By optimizing well construction costs and accelerating project delivery timelines, these innovations directly contribute to improved operational efficiency and reduced carbon footprint, aligning with both PDO's corporate strategy and Oman Vision 2040.

The success achieved in PDO's Northern Oman gas fields illustrates how customized drilling fluids, vibration mitigation tools, and advanced completion techniques such as multi-stage fracturing can collectively transform performance outcomes. However, the real value lies not only in the adoption of these technologies but also in the systematic approach to continuous improvement. Embedding robust after-action reviews into the planning and design phase ensures that lessons learned are captured, standardized, and translated into best practices across future operations.

Ultimately, these initiatives highlight the importance of integrating innovation with structured learning processes. By sustaining this cycle of trial, evaluation, and refinement, PDO and the wider industry can progressively overcome technical challenges, enhance project economics, and contribute meaningfully to the broader sustainability agenda.

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